

Below is a complete, priority-ordered launch playbook. Work top-to-bottom — every item before it is a dependency for the ones after.

Phase 0 — Fix the foundation (do this before anything else)

1. Populate your data before you market. Right now `/papers` shows **0 papers indexed**, which means you're asking friends to trust an empty shell. Before any launch, ingest at least 3–5 years of PYQs for **B.Tech CSE sem 3–6** (the most-searched segment), extract and cluster repeated questions, and tag each question with: subject code, unit, year(s) appeared, marks, type (theory/numerical/diagram). Your entire product promise (“know what to skip”) collapses if the dataset is thin. RyzenStudy's credibility comes from 1,447+ real papers — you don't need that many, but you need enough that a random B.Tech 4th-sem student searching “DBMS important questions AKTU” actually finds their subject. 【turn0search10】

2. Pick a wedge, not a buffet. Don't launch with “all AKTU courses” — launch with “**B.Tech 4th semester CSE, AKTU 2022–2026**” as your headline. A narrow, complete, deeply-analyzed dataset beats a broad, shallow one. Once your wedge users (4th-sem CSE students) love it, expand to B.Pharm and MCA in v2. This is the classic Facebook / Amazon playbook: dominate one niche, then expand.

Phase 1 — Technical foundation (latency, hosting, database)

3. Move off the vercel.app subdomain to a custom domain. Buy something short and brandable — `aktupriority.com`, `pyqrank.com`, `aktu.top`, or similar (~₹800/year on Cloudflare or Namecheap). A `*.vercel.app` URL caps your SEO, looks untrustworthy in a WhatsApp forward, and prevents you from building brand equity. Do this **before** you seed the link anywhere, or you'll be stuck migrating later.

4. Keep Vercel for the app, but front it with Cloudflare. Vercel's free tier is fine for your use case, but Cloudflare's edge network has more PoPs in India and lower TTFB for Indian users than Vercel's default routing. Put Cloudflare in front (free plan): your domain → Cloudflare CDN → Vercel origin. This single change shaves 100–300ms for UP students on Jio/Airtel and gives you free DDoS protection, caching rules, and a way to serve `robots.txt` / `sitemap.xml` directly at the edge without hitting your Next.js server. 【turn1search11】 【turn1search14】

5. Choose Supabase free tier for structured data, but design for caching. You have two data shapes: (a) normalized questions/subjects/units — perfect for Postgres (Supabase free tier gives

you 500MB DB, 50k monthly active users, enough for launch), and (b) heavy aggregated views like “top 20 repeated questions in DBMS Unit 3” — these should be **precomputed** and cached, not computed per-request. Use Next.js ISR (Incremental Static Regeneration) with a 24-hour revalidate for all list pages. Users on free Supabase report 2–5s cold starts — ISR means your pages serve from Vercel’s CDN as static HTML and only hit Supabase on revalidation or user-specific queries. 【turn0search5】 【turn0search9】

6. Cache aggressively at three layers. Layer 1: Cloudflare edge cache for static assets and robots.txt / sitemap.xml (Cache-Control: public, max-age=86400). Layer 2: Next.js ISR for every question-cluster page. Layer 3: In-memory (or Upstash Redis free tier) for any dynamic query like “top questions in this subject.” Target: first contentful paint under 1.5s on a mid-range Android over 4G. Test with PageSpeed Insights on a “Slow 4G” throttle — that’s your real user.

7. Build a PWA with offline cache. Students revise in hostels, libraries, and exam halls with bad connectivity. Add next-pwa, cache the app shell and the question pages a user visits, and let them “Save this subject for offline” — this is a killer feature RyzenStudy doesn’t have. Service workers let you cache pages so a user can open your app in airplane mode during a last-minute revision and still see their saved question lists. 【turn1search5】 【turn1search7】

Phase 2 — SEO & discoverability (this is how students will actually find you)

8. Keyword strategy: target the long tail, not the head. Don’t try to rank for “AKTU PYQ” — RyzenStudy already owns that. Target queries like “DBMS most repeated questions AKTU”, “Computer Networks unit 3 important questions”, “Theory of Automata repeated questions”, “B.Pharm 2nd sem most asked questions”. These are exactly what your product answers, they have lower competition, and they convert far better because the searcher has explicit intent. Use Google’s autocomplete and the “People also ask” box to expand your list.

9. Every subject × unit × question-cluster gets its own URL. This is non-negotiable. Structure: /btech/cse/sem-4/dbms/unit-3/repeated-questions, /bpharm/sem-2/most-asked-questions, /mca/sem-3/data-structures/top-questions. Each page needs a unique <title> (e.g., “DBMS Unit 3 Most Repeated Questions | AKTU B.Tech 2022–2026”) and <meta description> (e.g., “Every DBMS Unit 3 question clustered and ranked by how many times AKTU has asked it. Know what to prepare and what to skip.”). This is your SEO surface area — 50 subjects × 5 units × 3 question types = 750+ indexable pages from one dataset.

10. robots.txt — keep it simple. Serve this at /robots.txt:

```
User-agent: *  
Allow: /  
Disallow: /api/  
Sitemap: https://yourdomain.com/sitemap.xml
```

That's it. Don't block anything else. Test it in Google Search Console's robots.txt tester after deploy.

11. sitemap.xml — generate it dynamically. Use Next.js's built-in `app/sitemap.ts` (App Router) or `pages/sitemap.xml.js` to auto-generate a sitemap that lists every subject page, every unit page, and your 20–30 most-visited question clusters. Split into a sitemap index if you go past 5,000 URLs. Submit it to Google Search Console the day you deploy. This single step is the difference between Google finding your pages in 2 days vs. 2 months.

12. Add JSON-LD structured data. On each question-cluster page, add `FAQPage` schema (question text as the Q, “asked N times in years X, Y, Z” as the A) and `BreadcrumbList` schema. This makes your pages eligible for FAQ rich results and AI Overview citations — which is where a growing chunk of student searches end up now. 【turn0search1】

13. Internal linking. Every question page links to: its sibling questions in the same unit, the parent subject page, and 3–5 “students also searched for” related subjects. This passes PageRank through your site and keeps users browsing (reducing bounce, which feeds back into SEO).

Phase 3 — Analytics, trust, and legal

14. Wire up analytics on day one. Install Vercel Analytics (built-in, privacy-friendly) + Google Search Console (non-negotiable for SEO) + a lightweight event tracker like Plausible or PostHog free tier to see which subjects/users are sticky. Without this you're flying blind — you won't know if B.Pharm students bounce or if your DBMS page is your traffic magnet.

15. Add trust pages. Students (and Google) distrust thin sites. Create: `/about` (your face, your AKTU branch, your story — “I built this because I wasted 3 hours before my DBMS exam”), `/privacy` (a 3-paragraph plain-English policy — use a generator if needed), `/disclaimer` (“Not affiliated with AKTU. Questions sourced from publicly available past papers.”), and `/contact` (a simple form or a Telegram link). RyzenStudy's About page with a real student founder photo is a big trust signal — copy that pattern. 【turn0search10】

16. Register for Google Search Console and Bing Webmaster Tools. Submit your sitemap, request indexing on your top 20 pages, and monitor for crawl errors weekly. This is your SEO dashboard.

Phase 4 — Sharing & distribution (where most devs fail)

17. Make every question-cluster page natively shareable. Add a WhatsApp share button (use `https://wa.me/?text=...` with a pre-filled message like “DBMS Unit 3 most repeated questions — 6 of these have appeared 4+ times 📌 [link]”) and a native Web Share API button on mobile. The share text should sell the insight, not just the link. When a student forwards this to their class WhatsApp group, the text does the pitching — not the URL.

18. Generate shareable images. Use Vercel’s `@vercel/og` (ImageResponse) to auto-generate an OpenGraph image per subject page — e.g., a clean card showing “DBMS — Top 10 Repeated Questions · AKTU B.Tech Sem 4 · 2022–2026” with your logo. When someone pastes your link in WhatsApp/Telegram, this image is what makes people tap. A text-only link gets ignored; a designed card gets clicked.

19. Seed it in AKTU Telegram groups. There are dozens of large AKTU student Telegram groups for notes, PYQs, and exam discussion — some with 5–10k members. 【turn0search15】

【turn0search16】 Join 10–15 of them. Don’t spam your link on day one. Spend a week answering questions (“Which book for Theory of Automata?”), then post: “Made a free tool that shows which DBMS questions AKTU repeats — covered 2022–2026, no login. Feedback welcome: [link]”. Answer follow-ups personally. This is your highest-ROI acquisition channel — it costs nothing and reaches your exact target.

20. Launch on Reddit r/AKTU. A post like “I built a free tool that ranks AKTU PYQ questions by how often they repeat — sem 4 CSE covered, more coming” with a 30-second Loom/GIF demo will do numbers. Respond to every comment. The r/AKTU community has already shown it vibecodes and shares AKTU tools. 【turn0search7】

21. Recruit 5 “college ambassadors.” Pick one student from 5 different AKTU-affiliated colleges (your friends’ friends). Give them early access, ask them to share in their class group, and feature them on your About page. Students trust students from their own college more than any ad.

22. Add a “Made by [Your Name], [Your College]” badge on every page. Personal brands travel better than anonymous tools. When someone shares your link, “built by a CSE student at [college]” is the credibility hook that makes their friend click.

Phase 5 — Launch sequence

23. Soft launch to 20 friends first (week 1). Send the link personally to 20 AKTU students across different branches/semesters. Ask three questions after 48 hours: “What did you search for?”, “Did

you find it?”, “What’s missing?”. Fix the top 3 complaints. This is your free QA.

24. Then seed to 5 Telegram groups + r/AKTU (week 2). Same message, personalized per group. Track which channel brings users who actually browse 3+ pages (use PostHog). Double down on the winning channel.

25. Public launch (week 3). Post on LinkedIn, Twitter/X with a demo GIF, and submit to Product Hunt (optional but free backlinks). Keep the story personal: *“I’m a B.Tech student at [college] and I built this because...”* — not “Introducing my SaaS.”

Phase 6 — Post-launch loop

26. Set up a feedback widget on every page. A single line at the bottom: *“Was this useful? 👍 / 👎 / [suggest a fix]”* — pipe responses to a Google Sheet. Reply to every piece of feedback within 24h. This is how you turn users into evangelists.

27. Ship one visible improvement per week. “Added B.Pharm sem 3”, “Added MCA sem 2”, “Added offline mode for DBMS” — post each update in the same Telegram groups. Consistency is what separates a tool that grows from a tool that dies.

28. Expand to B.Pharm and MCA only after B.Tech CSE sem 3–6 is genuinely good. Resist the temptation to go wide. A B.Pharm student who lands on an empty B.Pharm page never comes back.

The one-paragraph summary

Get real data in first, move to a custom domain, put Cloudflare in front of Vercel for Indian latency, generate 500+ SEO pages (subject × unit × question-type) with ISR caching and JSON-LD, submit the sitemap to Search Console, build a PWA for offline revision, add WhatsApp-native share buttons with auto-generated OG images, seed it in AKTU Telegram groups and r/AKTU with a personal story (not a pitch), recruit 5 college ambassadors, soft launch to 20 friends, then public launch in week 3, and ship one visible improvement every week based on a feedback widget on every page.

Your product’s unfair advantage isn’t the papers — RyzenStudy has more. It’s **the analysis layer**. Every piece of advice above is designed to make sure a student who’s googling “DBMS important questions the night before my exam” finds *you* first, clicks in under a second, gets their answer in 10 seconds, and forwards it to their class group before closing the tab.